**PHIVOLCS Special Report –** Ms6.9 Negros Oriental Earthquake (February 6, 2012)

**Epicenter:** At 9.97°N, 123.14°E (5 km Northwest of Tayasan, Negros Oriental)

| Reported Earthquake Intensity* | Province | City / Municipality |
|---|---|---|
| VII (destructive) | Negros Oriental | La Libertad<br>Guihulngan City<br>Tayasan<br>Vallehermoso |

| Places | Coordinates | Findings (from the Method) |
|---|---|---|
| La Libertad, Negros Oriental | 10.0830° N, 123.1576° E | |
| Guihulngan City, Negros Oriental | 10.1832° N, 123.2139° E | |
| Tayasan, Negros Oriental | 9.9492° N, 123.0564° E | |
| Vallehermoso, Negros Oriental | 10.3391° N, 123.2925° E | |

## Intensity-Based Epicenter Estimate Report

### 1) Initial (Unfiltered) Epicenter Estimate

Epicenter (H, K): 11.074393° N, 121.567766° E

Epicenter 1σ interval: (H ± 0.398698°, K ± 0.724665°)

Distance to PHIVOLCS Measured Epicenter: 211.244 km

*Radii to (H, K)*

| Point | Radius (deg) | Radius (km) |
|---|---|---|
| P1 | 1.873615 | 205.791 |
| P2 | 1.871893 | 205.387 |
| P3 | 1.866036 | 205.282 |
| P4 | 1.874930 | 205.414 |

Minimum R: 1.866036° (~205.282 km)

Maximum R: 1.874930° (~205.414 km)

Mean R_prox: 1.871618°

Radius std: 0.003398°

### 2) Recalculated Epicenter (After Outlier Removal)

Epicenter (H, K): 11.074393° N, 121.567766° E

Epicenter 1σ interval: (H ± 0.398698°, K ± 0.724665°)

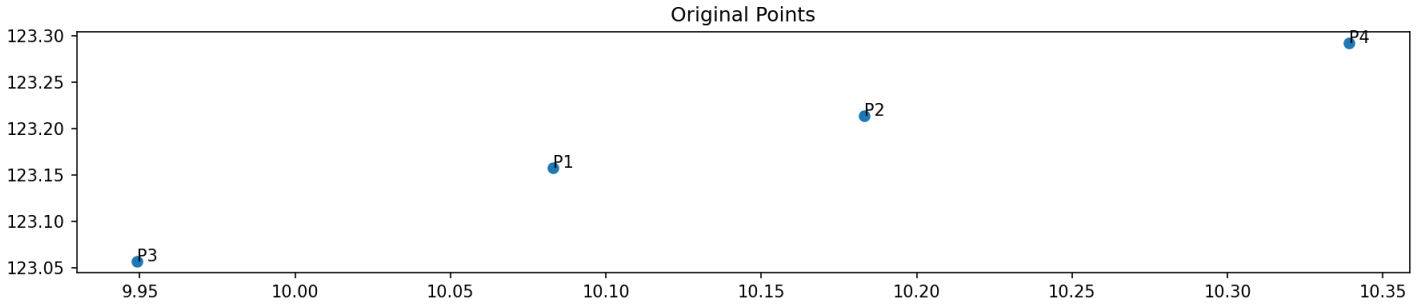
Distance to PHIVOLCS Measured Epicenter: 211.244 km

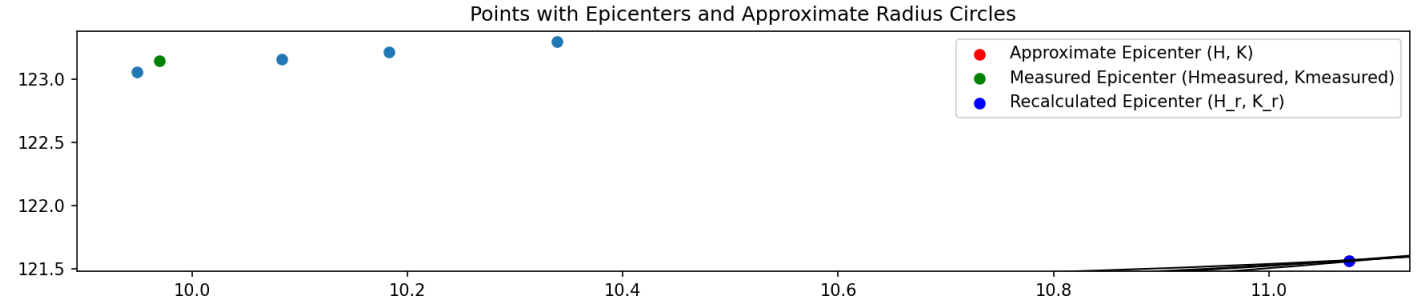
*Radii to (H, K)*

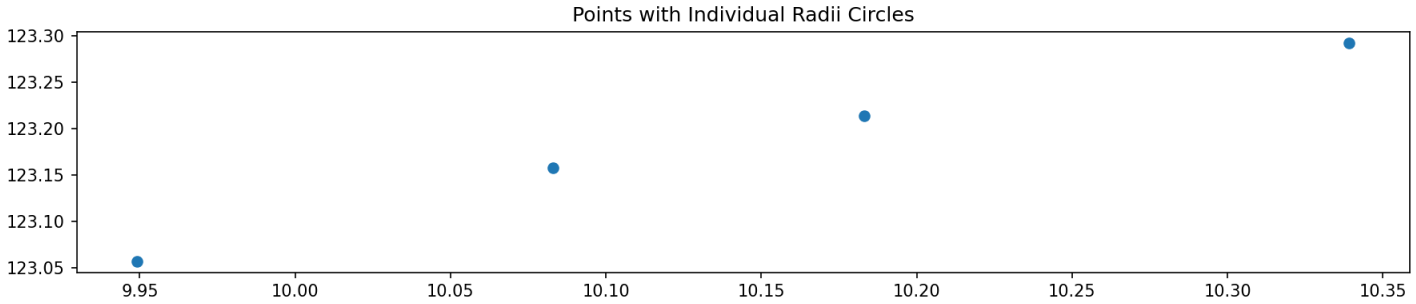
| Point | Radius (deg) | Radius (km) |
|---|---|---|
| P1 | 1.873615 | 205.791 |
| P2 | 1.871893 | 205.387 |
| P3 | 1.866036 | 205.282 |
| P4 | 1.874930 | 205.414 |

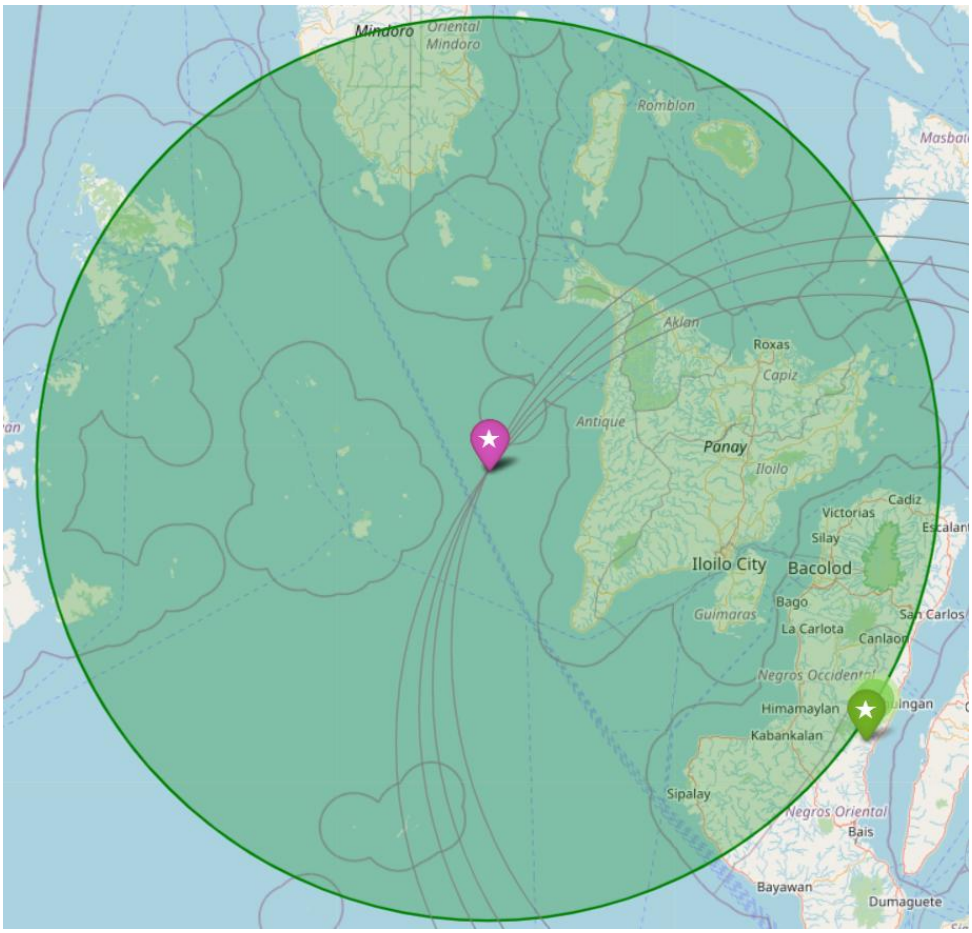
Minimum R: 1.866036° (~205.282 km)

Maximum R: 1.874930° (~205.414 km)


Original Points
P1
P2
P3
P4


Points with Epicenters and Approximate Radius Circles
Approximate Epicenter (H, K)
Measured Epicenter (Hmeasured, Kmeasured)
Recalculated Epicenter (H_r, K_r)


Points with Individual Radii Circles


Mindoro
Oriental Mindoro
Romblon
Aklan
Roxas
Capiz
Antique
Panay
Iloilo
Cadiz
Victorias
Silay
Iloilo City
Bacolod
Bago
Guimaras
La Carlota
Negros Occidental
Himamaylan
Kabankalan
Sipalay
Negros Oriental
Bais
Bayawan
Dumaguete

Intensity VI

| | | |
|---|---|---|
| **VI (very strong)** | Negros Oriental | Manjuyod<br>Tanjay<br>Dumaguete City |
| | Negros Occidental | La Carlota City<br>La Castellana |
| | Cebu | Argao<br>Barili<br>Cebu City<br>Dalaguete |
| | Bohol | Clarin |

| Places | Coordinates | Findings (from the Method) |
|---|---|---|
| Manjuyod, Negros Oriental | 9.7065° N, 123.0676° E | |
| Tanjay, Negros Oriental | 9.5149° N, 123.1580° E | |
| Dumaguete City, Negros Oriental | 9.2984° N, 123.2925° E | |
| La Carlota City, Negros Occidental | 10.4263° N, 122.9217° E | |
| La Castellana, Negros Occidental | 10.3320° N, 123.1014° E | |
| Argao, Cebu | 9.8811° N, 123.6053° E | |
| Barili, Cebu | 10.1165° N, 123.5087° E | |
| Cebu City, Cebu | 10.2926° N, 123.9022° E | |
| Dalaguete, Cebu | 9.7620° N, 123.5333° E | |
| Clarin, Bohol | 9.9452° N, 124.0307° E | |

## Intensity-Based Epicenter Estimate Report

### 1) Initial (Unfiltered) Epicenter Estimate

Epicenter (H, K): 9.233100° N, 123.242759° E
Epicenter 1σ interval: (H ± 7.906629°, K ± 4.287532°)
Distance to PHIVOLCS Measured Epicenter: 82.710 km

***Radii to (H, K)***

| Point | Radius (deg) | Radius (km) |
|---|---|---|
| P1 | 0.504765 | 56.036 |
| P2 | 0.294271 | 32.685 |
| P3 | 0.082086 | 9.084 |
| P4 | 1.235639 | 137.262 |
| P5 | 1.107954 | 123.170 |
| P6 | 0.742522 | 82.293 |
| P7 | 0.922561 | 102.464 |
| P8 | 1.247959 | 138.208 |
| P9 | 0.603447 | 66.888 |
| P10 | 1.062044 | 117.188 |

Minimum R: 0.082086° (~9.084 km)
Maximum R: 1.247959° (~138.208 km)
Mean R_prox: 0.780325°
Radius std: 0.382801°

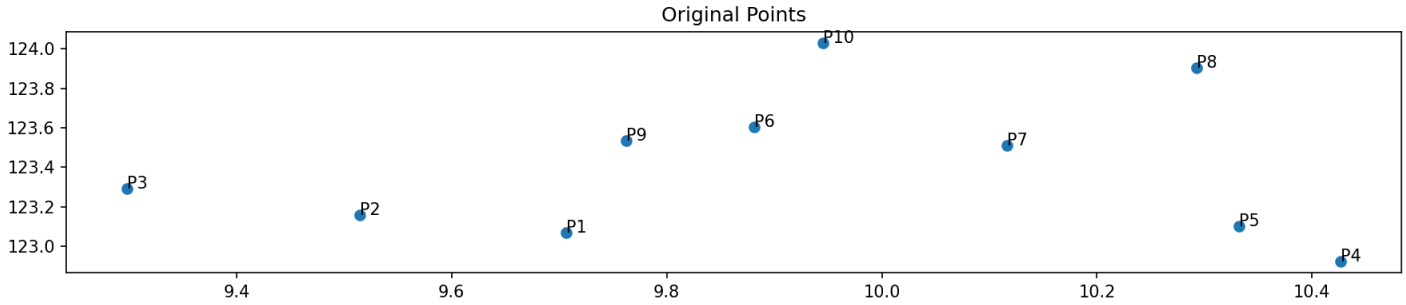
### 2) Recalculated Epicenter (After Outlier Removal)

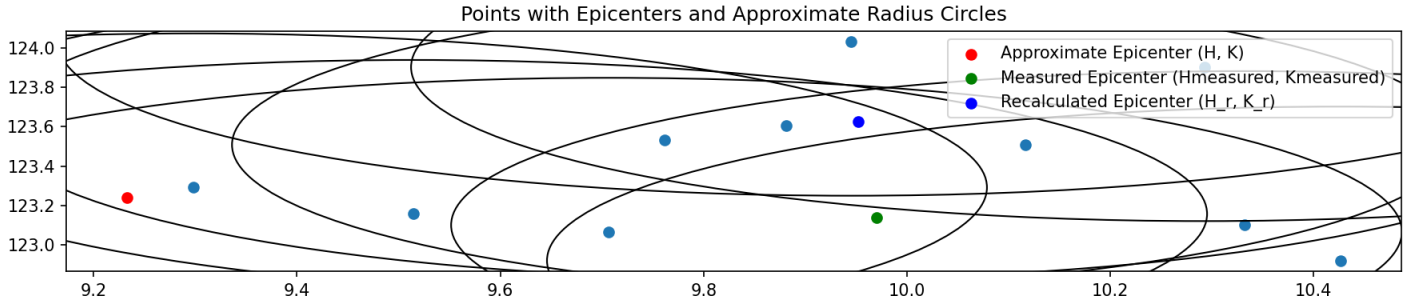
Epicenter (H, K): 9.952039° N, 123.627029° E
Epicenter 1σ interval: (H ± 1.007664°, K ± 0.904347°)
Distance to PHIVOLCS Measured Epicenter: 53.376 km

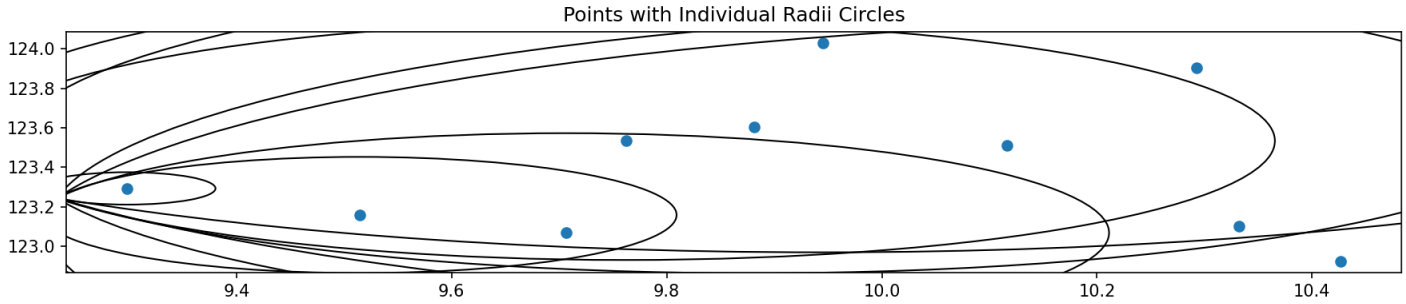
***Radii to (H, K)***

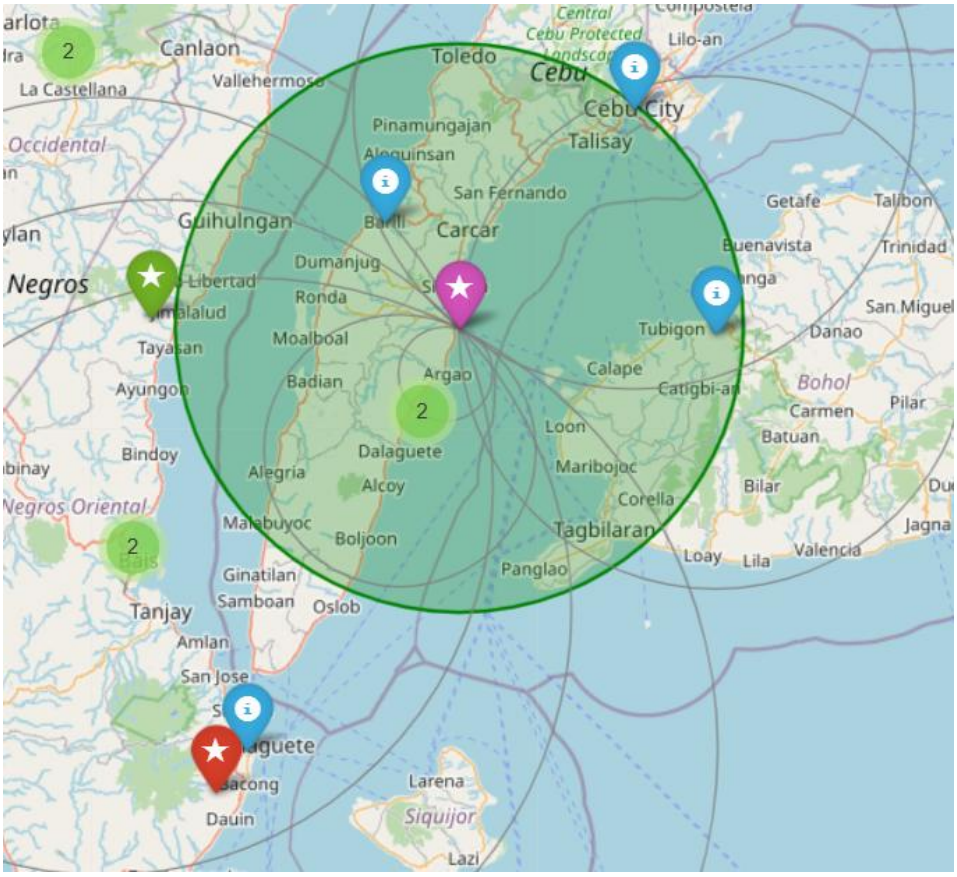
| Point | Radius (deg) | Radius (km) |
|---|---|---|
| P1 | 0.610942 | 67.099 |
| P2 | 0.641154 | 70.746 |
| P3 | 0.734271 | 81.410 |
| P6 | 0.074192 | 8.239 |
| P8 | 0.437837 | 48.387 |
| P9 | 0.211896 | 23.494 |
| P10 | 0.403729 | 44.218 |

Minimum R: 0.074192° (~8.239 km)
Maximum R: 0.734271° (~81.410 km)
Mean R_prox: 0.444860°
Radius std: 0.221450°


Original Points
P10
P8
P9
P6
P7
P3
P2
P1
P5
P4
124.0
123.8
123.6
123.4
123.2
123.0
9.4
9.6
9.8
10.0
10.2
10.4


Points with Epicenters and Approximate Radius Circles
Approximate Epicenter (H, K)
Measured Epicenter (Hmeasured, Kmeasured)
Recalculated Epicenter (H_r, K_r)
124.0
123.8
123.6
123.4
123.2
123.0
9.2
9.4
9.6
9.8
10.0
10.2
10.4


Points with Individual Radii Circles
124.0
123.8
123.6
123.4
123.2
123.0
9.4
9.6
9.8
10.0
10.2
10.4


Canlaon
Toledo
Cebu
Lilo-an
Cebu City
Talisay
Pinamungajan
Aloguinsan
San Fernando
Carcar
Barili
Guihulngan
Dumanjug
Ronda
Moalboal
Tayasan
Ayungon
Badian
Argao
Dalaguete
Alegria
Alcoy
Boljoon
Malabuyoc
Ginatilan
Samboan
Oslob
Tanjay
Amlan
San Jose
Dauin
Siquijor
Larena
Lazi
Panglao
Tagbilaran
Loon
Maribojoc
Corella
Bilar
Calape
Tubigon
Catigbi-an
Getafe
Talibon
Buenavista
Trinidad
Danao
San Miguel
Carmen
Batuan
Pilar
Bohol
Loay
Lila
Valencia
Jagna
Negros
Negros Oriental
Occidental
La Castellana
Bindoy
Libertad
2

Intensity V

| | | |
|---|---|---|
| **V (strong)** | Aklan | Roxas City |
| | Capiz | Dao<br>Ivisan |
| | Iloilo | Iloilo City |
| | Negros Oriental | Ayungon<br>Canlaon City |
| | Negros Occidental | Bacolod City<br>Binalbagan<br>Candoni<br>Hinigaran<br>Sagay City<br>San Carlos City |
| | Guimaras | |
| | Bohol | Tagbilaran City |

| **Places** | **Coordinates** | **Findings (from the Method)** |
|---|---|---|
| Roxas City, Aklan | 11.5839° N, 122.7522° E | |
| Dao, Capiz | 11.3890° N, 122.7068° E | |
| Ivisan, Capiz | 11.5105° N, 122.6842° E | |
| Iloilo City, Iloilo | 10.7302° N, 122.5591° E | |
| Ayungon, Negros Oriental | 9.8718° N, 123.0564° E | |
| Canlaon City, Negros Oriental | 10.3605° N, 123.1914° E | |
| Bacolod City, Negros Occidental | 10.6676° N, 122.9456° E | |
| Binalbagan, Negros Occidental | 10.1951° N, 122.8651° E | |
| Candoni, Negros Occidental | 9.7611° N, 122.6504° E | |
| Hinigaran, Negros Occidental | 10.2748° N, 122.9212° E | |
| Sagay City, Negros Occidental | 10.8456° N, 123.3711° E | |
| Tagbilaran City | 9.6534° N, 123.8670° E | |
| Guimaras<br>San Carlos City | 10.5929° N, 122.6325° E<br>10.4826° N, 123.4194° E | |

# Intensity-Based Epicenter Estimate Report

## 1) Initial (Unfiltered) Epicenter Estimate

Epicenter (H, K): 10.816565° N, 123.862542° E

Epicenter 1σ interval: (H ± 2.846093°, K ± 7.398261°)

Distance to PHIVOLCS Measured Epicenter: 122.906 km

***Radii to (H, K)***

| Point | Radius (deg) | Radius (km) |
|---|---|---|
| P1 | 1.349690 | 148.150 |
| P2 | 1.289737 | 141.260 |
| P3 | 1.367492 | 149.927 |
| P4 | 1.306300 | 142.705 |
| P5 | 1.241953 | 137.157 |
| P6 | 0.811435 | 89.179 |
| P7 | 0.928964 | 101.533 |
| P8 | 1.175206 | 129.102 |
| P9 | 1.607264 | 177.089 |
| P10 | 1.086110 | 119.241 |
| P11 | 0.492299 | 53.769 |
| P12 | 1.163174 | 129.339 |
| P13 | 1.250212 | 136.676 |
| P14 | 0.554895 | 61.026 |

Minimum R: 0.492299° (~53.769 km)

Maximum R: 1.607264° (~177.089 km)

Mean R_prox: 1.116052°

Radius std: 0.304453°

## 2) Recalculated Epicenter (After Outlier Removal)

Epicenter (H, K): 10.635547° N, 123.253566° E

Epicenter 1σ interval: (H ± 0.901207°, K ± 2.139942°)

Distance to PHIVOLCS Measured Epicenter: 75.041 km

***Radii to (H, K)***

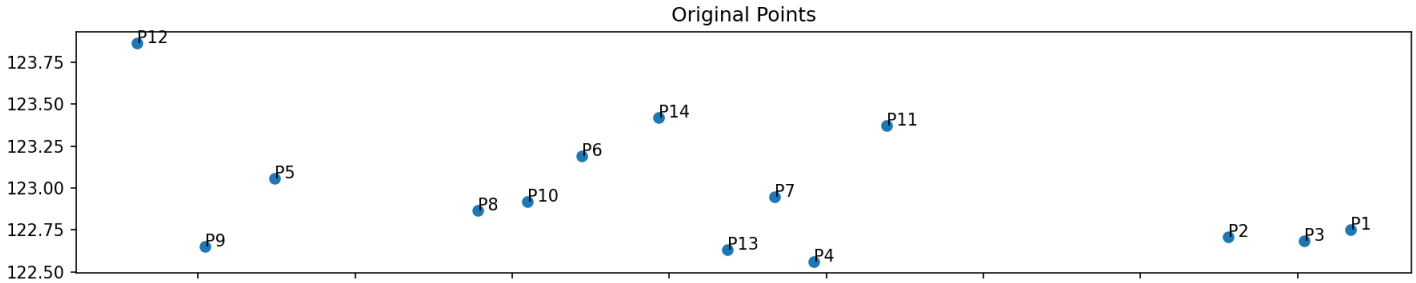
| Point | Radius (deg) | Radius (km) |
|---|---|---|
| P7 | 0.309629 | 33.842 |
| P9 | 1.062293 | 117.523 |
| P11 | 0.240700 | 26.654 |

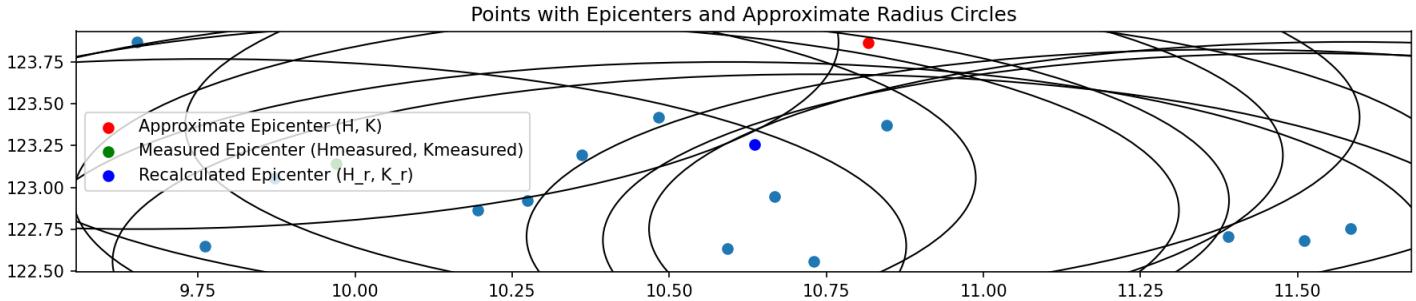
Minimum R: 0.240700° (~26.654 km)

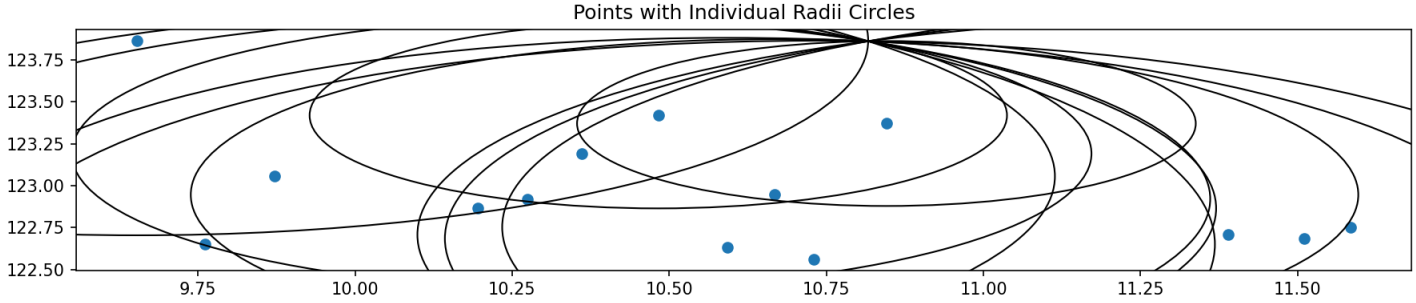
Maximum R: 1.062293° (~117.523 km)

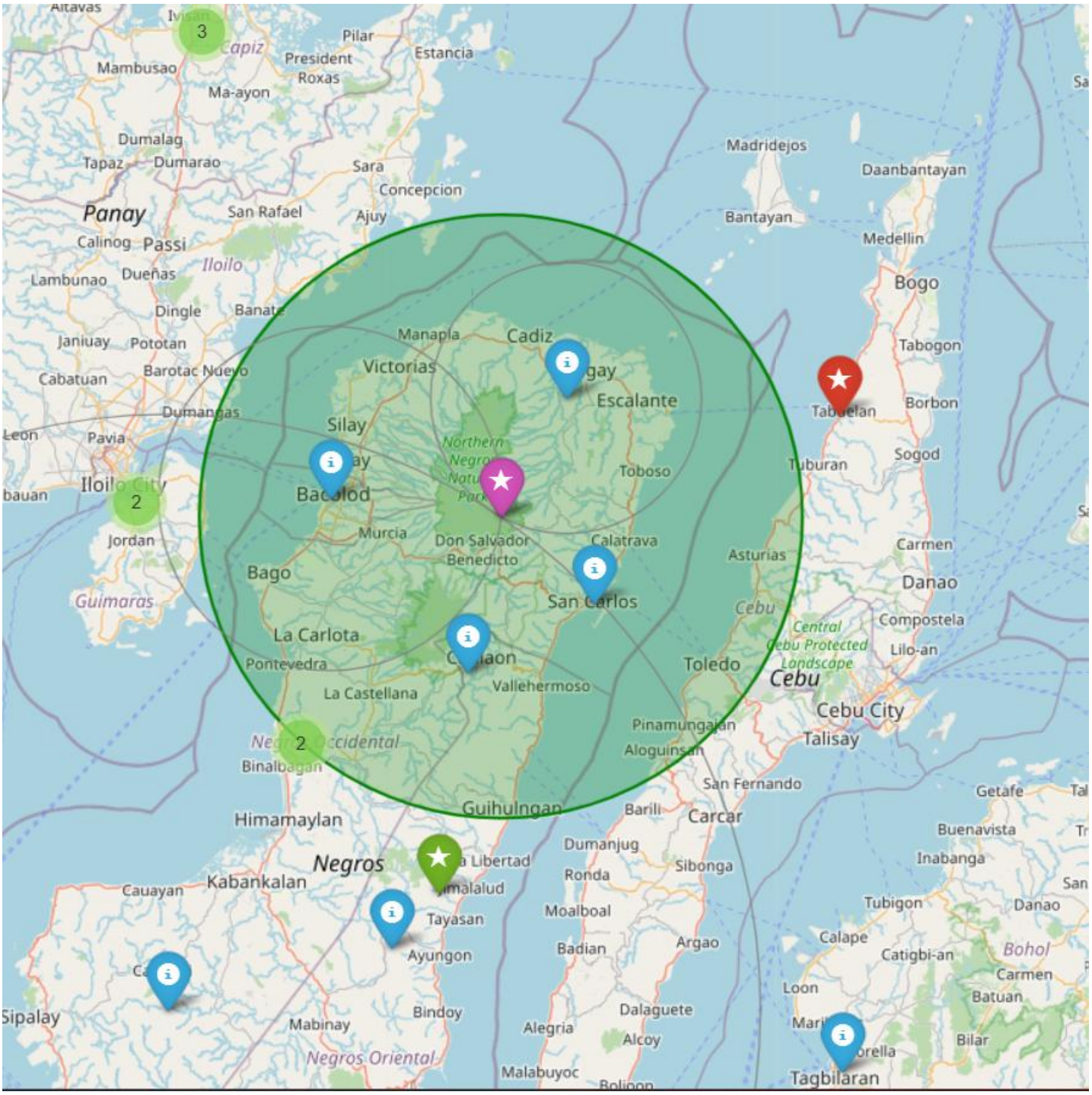
Mean R_prox: 0.537541°

Radius std: 0.372121°




Mambusao
Pilar
President Roxas
Estancia
Ma-ayon
Dumalag
Tapaz
Dumarao
Sara
Concepcion
Panay
San Rafael
Ajuy
Calinog
Passi
Iloilo
Lambunao
Dueñas
Dingle
Janiuay
Pototan
Cabatuan
Barotac Nuevo
Dumangas
Leon
Pavia
Iloilo City
Jordan
Guimaras
Manapla
Cadiz
Victorias
Escalante
Silay
Bacolod
Murcia
Don Salvador Benedicto
Calatrava
Toboso
Bago
San Carlos
La Carlota
Pontevedra
La Castellana
Valladolid
Himamaylan
Negros
Kabankalan
Cauayan
Tayasan
Ayungon
Sipalay
Mabinay
Bindoy
Negros Oriental
Guihulngan
Barili
Dumanjug
Ronda
Moalboal
Badian
Alegria
Alcoy
Malabuyoc
Madridejos
Bantayan
Daanbantayan
Medellin
Bogo
Tabogon
Tabuelan
Borbon
Tuburan
Sogod
Asturias
Carmen
Danao
Cebu
Central Cebu Protected Landscape
Compostela
Lilo-an
Toledo
Cebu City
Talisay
Pinamungajan
Aloguinsan
San Fernando
Carcar
Sibonga
Argao
Getafe
Buenavista
Inabanga
Tubigon
Calape
Catigbi-an
Bohol
Loon
Batuan
Dalaguete
Bilar
Tagbilaran

Intensity IV

| | | |
|---|---|---|
| **IV (moderately strong)** | Aklan | Banga<br>Balete<br>Batab<br>Kalibo<br>Makatao<br>New Washington<br>Numancia |
| | Antique | Anini-y<br>Pandan<br>Patnungon<br>San Jose de Buenavista |
| | Negros Occidental | Sipalay |
| | Leyte | Ormoc City |
| | Iloilo | Santa Barbara |
| | Zamboanga Del Norte | Dipolog City |

| **Places** | **Coordinates** | **Findings (from the Method)** |
|---|---|---|
| Banga, Aklan | 11.5756° N, 122.3221° E | |
| Balete, Aklan | 11.5302° N, 122.3674° E | |
| Batab, Aklan | 11.5473° N, 122.5486° E | |
| Kalibo, Aklan | 11.7073° N, 122.3699° E | |
| Makatao, Aklan | 11.7006° N, 122.2767° E | |
| New Washington, Aklan | 11.6466° N, 122.3901° E | |
| Numancia, Aklan | 11.7177° N, 122.3391° E | |
| Anini-y, Antique | 10.4515° N, 121.9587° E | |
| Pandan, Antique | 11.7518° N, 122.1065° E | |
| Patnungon, Antique | 10.9230° N, 122.0610° E | |
| San Jose de Buenavista, Antique | 10.7490° N, 121.9413° E | |
| Sipalay, Negros Occidental | 9.7198° N, 122.4920° E | |
| Ormoc City, Leyte | 11.0384° N, 124.6193° E | |
| Santa Barbara, Iloilo | 10.8271° N, 122.5327° E | |
| Dipolog City, Zamboanga Del Norte | 8.5872° N, 123.3404° E | |

## Intensity-Based Epicenter Estimate Report

### 1) Initial (Unfiltered) Epicenter Estimate

Epicenter (H, K): 11.155884° N, 122.594792° E

Epicenter 1σ interval: (H ± 5.507368°, K ± 7.497559°)

Distance to PHIVOLCS Measured Epicenter: 144.706 km

***Radii to (H, K)***

| Point | Radius (deg) | Radius (km) |
|---|---|---|
| P1 | 0.500522 | 55.334 |
| P2 | 0.437972 | 48.446 |
| P3 | 0.394132 | 43.814 |
| P4 | 0.595513 | 66.032 |
| P5 | 0.630791 | 69.790 |
| P6 | 0.531696 | 58.951 |
| P7 | 0.617264 | 68.405 |
| P8 | 0.949089 | 104.698 |
| P9 | 0.770418 | 84.985 |
| P10 | 0.582382 | 63.753 |
| P11 | 0.769809 | 84.478 |
| P12 | 1.439758 | 160.081 |
| P13 | 2.027914 | 221.292 |
| P14 | 0.334596 | 37.182 |
| P15 | 2.674709 | 297.073 |

Minimum R: 0.334596° (~37.182 km)

Maximum R: 2.674709° (~297.073 km)

Mean R_prox: 0.883771°

Radius std: 0.641644°

### 2) Recalculated Epicenter (After Outlier Removal)

Epicenter (H, K): 10.999540° N, 122.779913° E

Epicenter 1σ interval: (H ± 1.557267°, K ± 2.706815°)

Distance to PHIVOLCS Measured Epicenter: 121.061 km

***Radii to (H, K)***

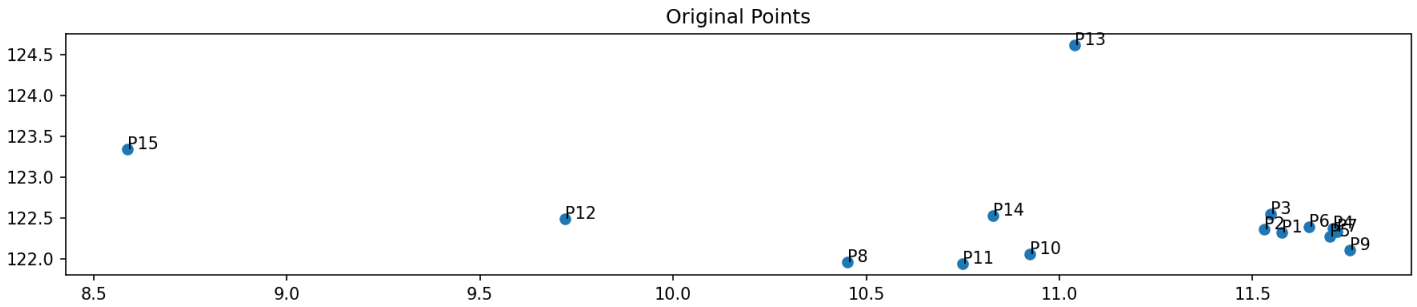
| Point | Radius (deg) | Radius (km) |
|---|---|---|
| P2 | 0.672136 | 74.199 |
| P10 | 0.722976 | 78.941 |

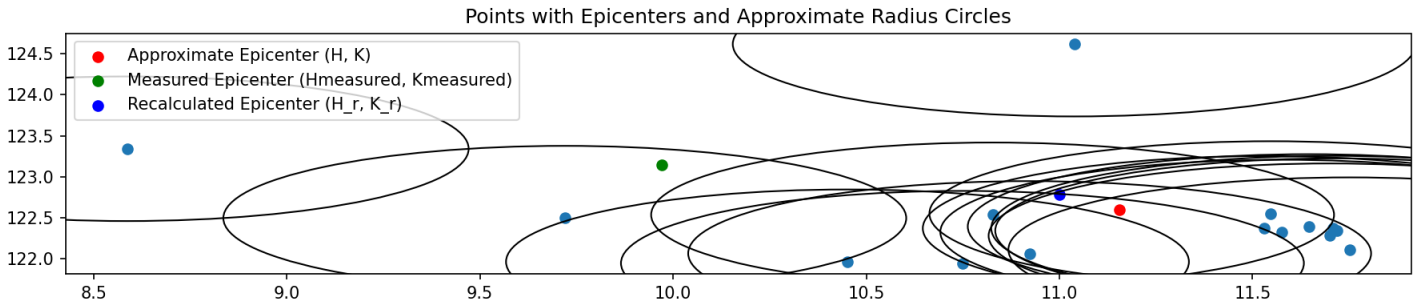
Minimum R: 0.672136° (~74.199 km)

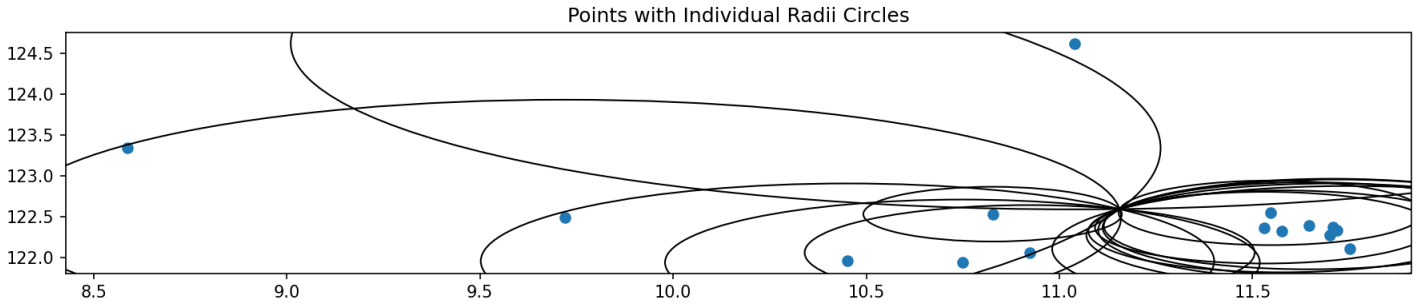
Maximum R: 0.722976° (~78.941 km)

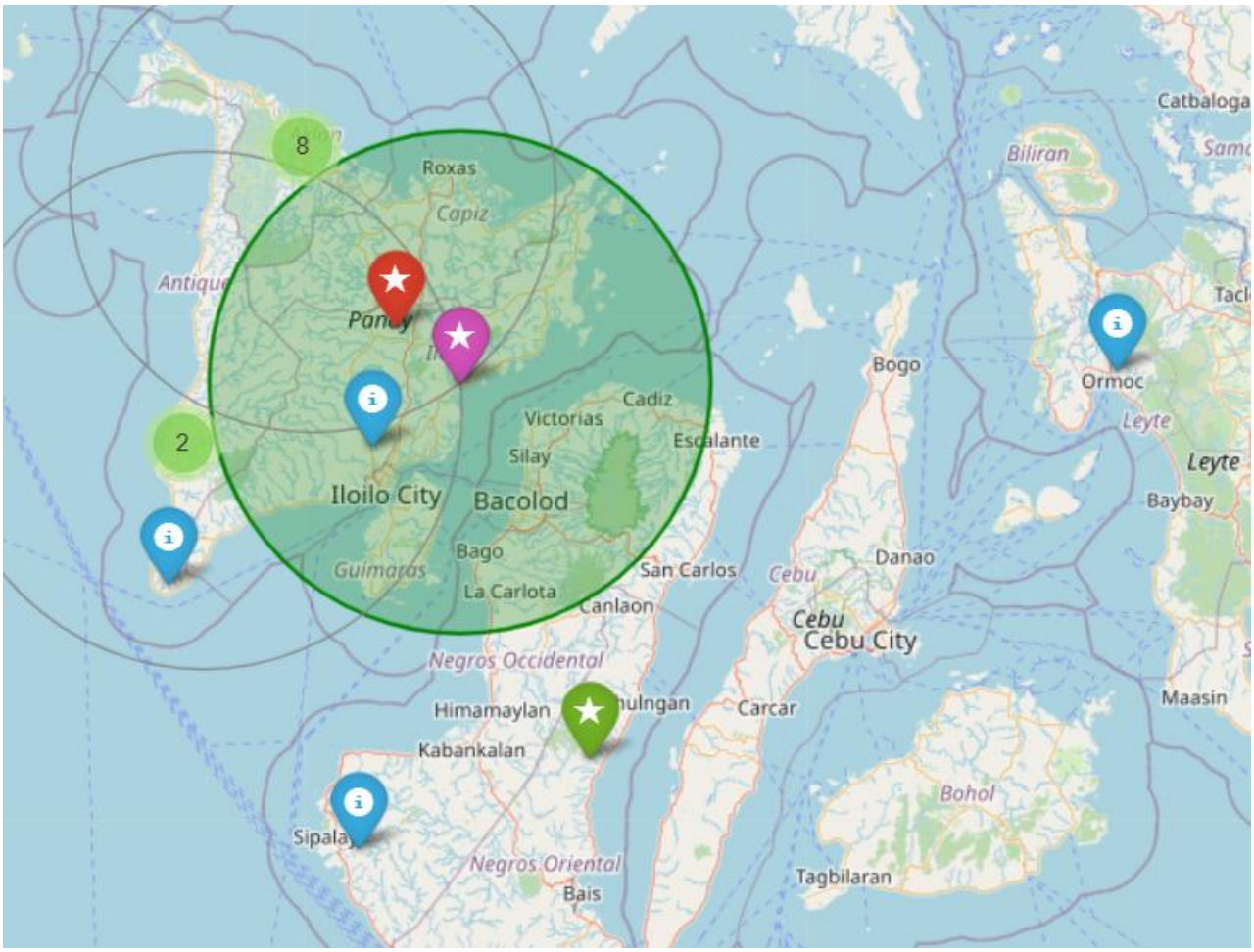
Mean R_prox: 0.697556°

Radius std: 0.025420°






Intensity III

| | | |
|---|---|---|
| **III (weak)** | Agusan Del Norte | Butuan City |
| | Albay | Legaspi City |
| | Misamis Oriental | Cagayan De Oro City |
| | Leyte | Tacloban City |
| | Samar | Catbalogan |
| | Southern Leyte | Saint Bernard |
| | Masbate | Masbate |

| **Places** | **Coordinates** | **Findings (from the Method)** |
|---|---|---|
| Butuan City, Agusan Del Norte | 8.9475° N, 125.5406° E | |
| Legaspi City, Albay | 13.1129° N, 123.7629° E | |
| Cagayan De Oro City, Misamis Oriental | 8.4803° N, 124.6498° E | |
| Tacloban City, Leyte | 11.2224° N, 124.9922° E | |
| Catbalogan City, Samar | 11.7752° N, 124.8828° E | |
| Saint Bernard, Southern Leyte | 10.3223° N, 125.1198° E | |
| Masbate, Masbate | 12.3717° N, 123.6219° E | |

# Intensity-Based Epicenter Estimate Report

## 1) Initial (Unfiltered) Epicenter Estimate

Epicenter (H, K): 11.547116° N, 123.726371° E

Epicenter 1σ interval: (H ± 16.231507°, K ± 17.177316°)

Distance to PHIVOLCS Measured Epicenter: 186.699 km

***Radii to (H, K)***

| Point | Radius (deg) | Radius (km) |
|---|---|---|
| P1 | 1.849945 | 201.613 |
| P2 | 1.803317 | 196.509 |
| P3 | 1.726549 | 188.120 |
| P4 | 1.410082 | 153.612 |
| P5 | 1.454919 | 161.665 |
| P6 | 1.621753 | 176.666 |
| P7 | 2.528519 | 280.451 |
| P8 | 3.671404 | 407.360 |

Minimum R: 1.410082° (~153.612 km)

Maximum R: 3.671404° (~407.360 km)

Mean R_prox: 2.008311°

Radius std: 0.706800°

## 2) Recalculated Epicenter (After Outlier Removal)

Epicenter (H, K): 11.031410° N, 123.439646° E

Epicenter 1σ interval: (H ± 5.757138°, K ± 6.539765°)

Distance to PHIVOLCS Measured Epicenter: 122.486 km

***Radii to (H, K)***

| Point | Radius (deg) | Radius (km) |
|---|---|---|
| P4 | 1.289619 | 141.297 |
| P5 | 2.022352 | 224.521 |
| P6 | 1.994958 | 217.885 |

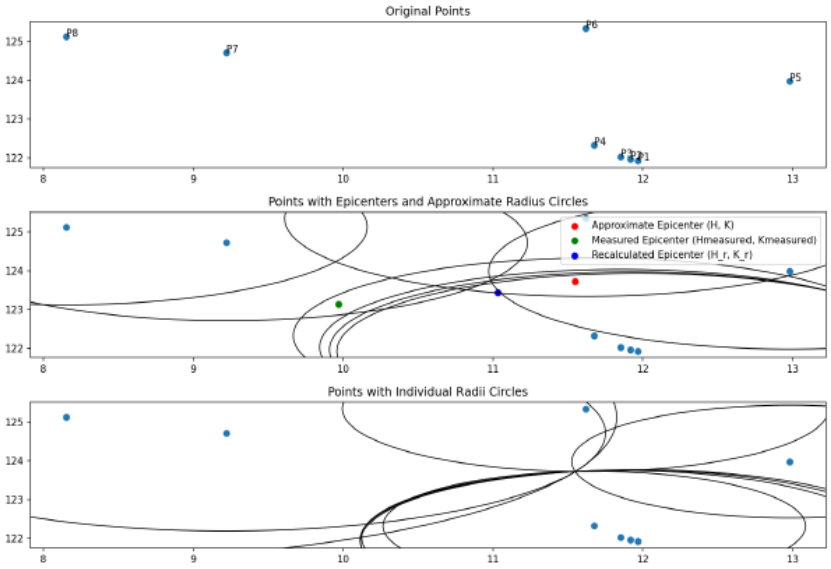
Minimum R: 1.289619° (~141.297 km)

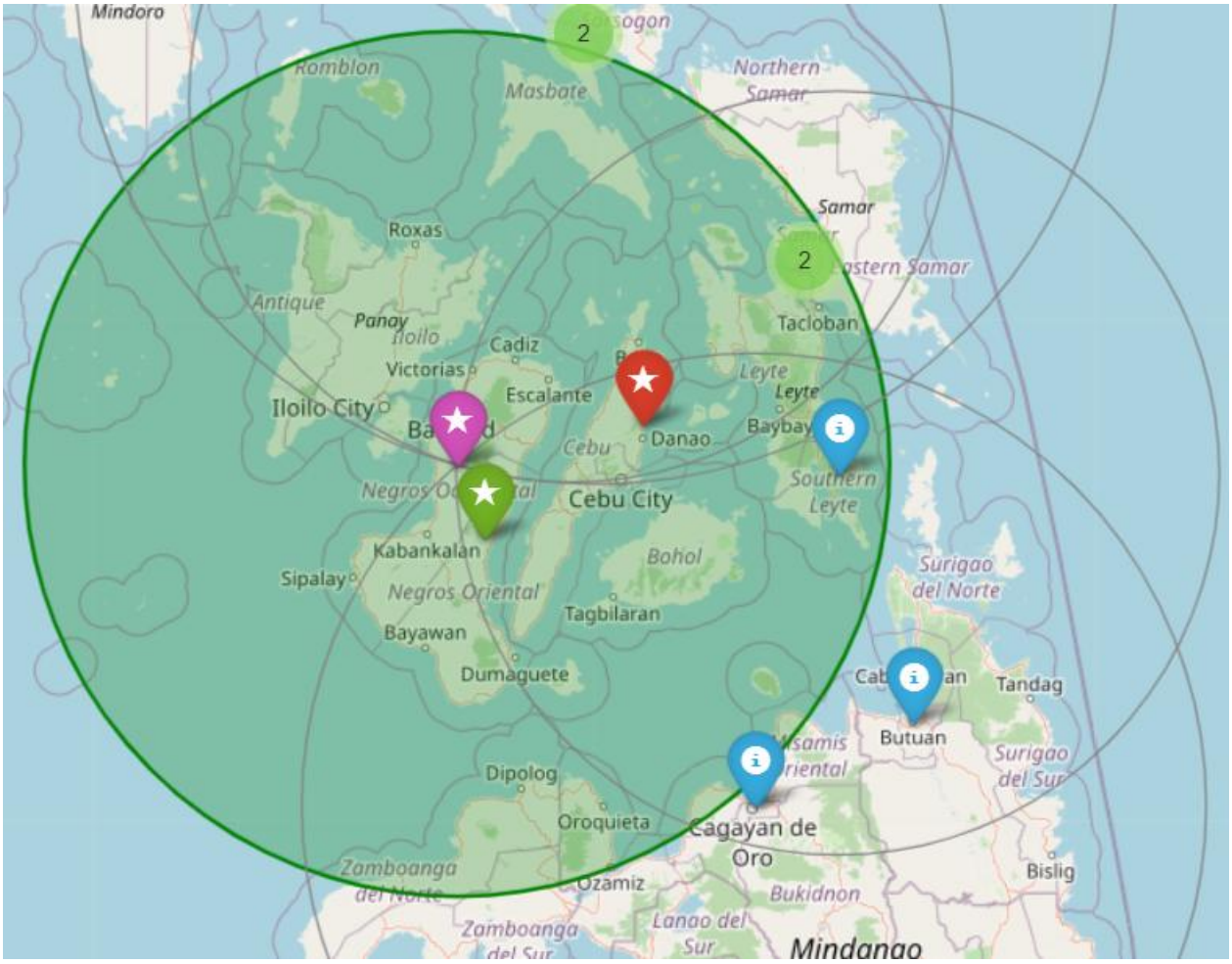
Maximum R: 2.022352° (~224.521 km)

Mean R_prox: 1.768976°

Radius std: 0.339141°

***Figure: Triplet Geometry and Radii***






Intensity II

| | | |
|---|---|---|
| **II (slightly felt)** | Aklan | Malay (Boracay)<br>Malay (Caticlan)<br>Nabas<br>Lezo |
| | Sorsogon | Sorsogon City (Cabid-an) |
| | Eastern Samar | Borongan |
| | Camiguin | Mambajao |
| | Bukidnon | Malaybalay |

| **Places** | **Coordinates** | **Findings (from the Method)** |
|---|---|---|
| Malay Boracay, Aklan | 11.9674° N, 121.9248° E | |
| Malay Caticlan, Aklan | 11.9175° N, 121.9615° E | |
| Nabas, Aklan | 11.8517° N, 122.0269° E | |
| Lezo, Aklan | 11.6750° N, 122.3221° E | |
| Sorsogon City, Sorsogon | 12.9789° N, 123.9848° E | |
| Borongan, Eastern Samar | 11.6174° N, 125.3466° E | |
| Mambajao, Camiguin | 9.2215° N, 124.7188° E | |
| Malaybalay, Bukidnon | 8.1540° N, 125.1285° E | |

# Intensity-Based Epicenter Estimate Report

## 1) Initial (Unfiltered) Epicenter Estimate

Epicenter (H, K): 10.598811° N, 124.026937° E

Epicenter 1σ interval: (H ± 1.929800°, K ± 7.462636°)

Distance to PHIVOLCS Measured Epicenter: 119.605 km

***Radii to (H, K)***

| Point | Radius (deg) | Radius (km) |
|---|---|---|
| P1 | 2.240090 | 247.439 |
| P2 | 2.527916 | 281.027 |
| P3 | 2.208178 | 245.269 |
| P4 | 1.149172 | 126.156 |
| P5 | 1.454782 | 160.706 |
| P6 | 1.127301 | 123.393 |
| P7 | 1.818568 | 202.016 |

Minimum R: 1.127301° (~123.393 km)

Maximum R: 2.527916° (~281.027 km)

Mean R_prox: 1.789430°

Radius std: 0.518926°

## 2) Recalculated Epicenter (After Outlier Removal)

Epicenter (H, K): 10.370993° N, 122.982738° E

Epicenter 1σ interval: (H ± 1.420250°, K ± 4.377966°)

Distance to PHIVOLCS Measured Epicenter: 47.795 km

***Radii to (H, K)***

| Point | Radius (deg) | Radius (km) |
|---|---|---|
| P2 | 2.850738 | 316.494 |
| P3 | 2.520677 | 278.632 |
| P6 | 2.137617 | 233.829 |
| P7 | 2.100324 | 233.124 |

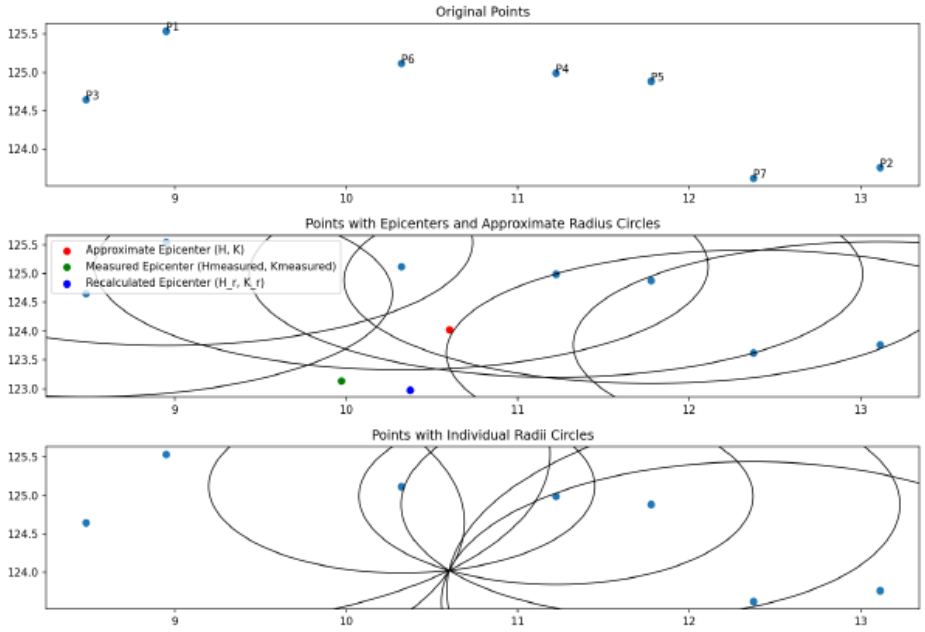
Minimum R: 2.100324° (~233.124 km)

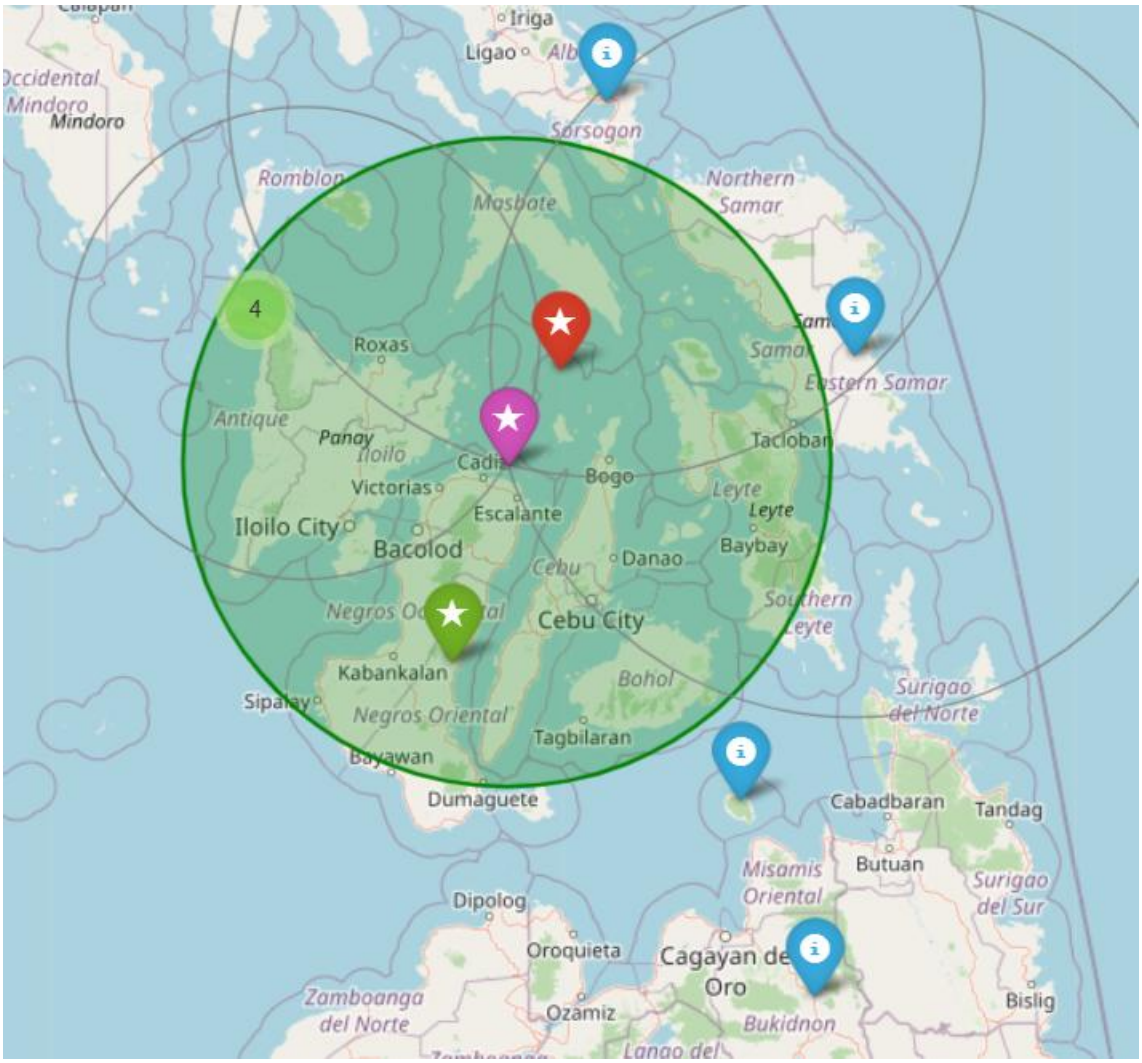
Maximum R: 2.850738° (~316.494 km)

Mean R_prox: 2.402339°

Radius std: 0.306740°

***Figure: Triplet Geometry and Radii***




Iriga
Ligao
Albay
Occidental Mindoro
Mindoro
Sorsogon
Romblon
Masbate
Northern Samar
4
Roxas
Samar
Eastern Samar
Antique
Panay
Iloilo
Cadiz
Bogo
Tacloban
Victorias
Leyte
Escalante
Iloilo City
Bacolod
Baybay
Cebu
Danao
Negros Occidental
Cebu City
Southern Leyte
Kabankalan
Bohol
Surigao del Norte
Sipalay
Negros Oriental
Tagbilaran
Bayawan
Dumaguete
Cabadbaran
Tandag
Butuan
Misamis Oriental
Surigao del Sur
Dipolog
Oroquieta
Cagayan de Oro
Zamboanga del Norte
Ozamiz
Bukidnon
Bislig
Lanao del

| CONSOLIDATED TABLE | | | | |
|---|---|---|---|---|
| Intensity | Initial Epicenter | Recalculated Epicenter | Outliers | Distance Error |
| Actual Location | 9.97° N, 123.14° E | - | - | - |
| VII | 11.0744° N, 121.5678° E | 11.0744° N, 121.5678° E | None | 211.244 km |
| VI | 9.2331° N, 123.2428° E | 9.9520° N, 123.6270° E | La Carlota City, La Castella Negros Oriental, and Barili, Cebu | 53.38 km |
| V | 10.8166° N, 123.8625° E | 10.6355° N, 123.2536° E | Numancia, Aklan Pandan, Antique San José de Buenavista, Antique | 75.041 km |
| IV | 11.1559° N, 122.5948° E | 10.9995° N, 122.7799° E | Only Balete, Aklan, and Patnungon, Antique seems stable | 121.061 km |
| III | 10.5988° N, 124.0269° E | 10.3701° N, 122.9827° E | Butuan City, Agusan del Norte Tacloban City, Leyte Catbalogan City, Samar | 47.795 km |
| II | 11.5471° N, 123.7264° E | 11.0314° N, 123.4396° E | Lezo, Aklan, Sorsogon City and Sorsogon, Borongan Eastern, Samar | 122.486 km |

**INTERPRETATION NARRATIVE BY INTENSITY LEVEL**

*(Negros Oriental — 06 Feb 2012, Mw 6.9; instrumental epicenter ≈ 9.97° N, 123.20° E)*

**INTENSITY VII (Destructive)**

**Provinces covered:** Negros Oriental
**Towns included:** Guihulngan City, La Libertad, Jimalalud, Tayasan, Ayungon, Bindoy, Manjuyod.

**Interpretation**
The VII reports line up along the **northeastern Negros Oriental coastline**, forming a near-collinear strip with poor azimuthal spread. Even though the shaking is near-source and severe (landslides, collapses), the geometry is unfavorable for radius intersections, so the inversion becomes directionally unstable.

**After filtering / result**

- **Recalculated Epicenter: 11.0744° N, 121.5678° E**
- **Distance from PHIVOLCS epicenter: ≈ 211 km**

**Assessment**
Large error arises from **coastal collinearity**, not from the method per se. VII marks severe damage reliably, but with this spatial layout it **cannot** triangulate the source.

**INTENSITY VI (Very Strong)**

**Provinces covered:** Negros Oriental, Negros Occidental, Cebu
**Towns included (sample):** Bais City, Tanjay City, Amlan, San Jose, Sibulan, Dumaguete City, Bacong, Valencia, Canlaon City; La Carlota City (Negros Occ.); Toledo City (Cebu).

**Interpretation**
VI spans multiple directions and substrates (volcanic slopes, coastal/alluvial plains, uplifted limestones, the Cebu block). This **multi-azimuth** spread contains source signal but introduces site/path heterogeneity, so radii scatter more than in well-balanced mainland sets.

**After filtering / result**

- **Recalculated Epicenter: 9.9520° N, 123.6270° E**
- **Distance from PHIVOLCS epicenter: ≈ 53.38 km**

**Assessment**
A **useful** band for this event: geometry improves vs VII (added azimuths), yielding a moderate misfit consistent with cross-island/heterogeneous geology.

**INTENSITY V (Strong)**

**Provinces covered:** Negros Oriental, Cebu, Siquijor, Bohol
**Towns included (sample):** Cebu City, Mandaue, Lapu-Lapu, Talisay, Minglanilla, Carcar (Cebu); Tagbilaran (Bohol); towns in Siquijor; western/central Negros cities.

**Interpretation**
V-level shaking forms a **multi-island belt** across Central Visayas. Basin amplification (Cebu coastal plain/reclaimed zones) and efficient propagation through straits expand the felt area while degrading geometric fidelity.

**After filtering / result**

- **Recalculated Epicenter: 10.6355° N, 123.2536° E**
- **Distance from PHIVOLCS epicenter: ≈ 75.041 km**

**Assessment**
Good for **regional strong-motion mapping**, but its mixed island geology and longer paths make epicentral convergence **moderate to weak**.

**INTENSITY IV (Moderately Strong)**

**Provinces covered:** Negros Oriental/Occidental, Cebu, Bohol, Siquijor, Iloilo, Capiz
**Towns included (sample):** Roxas City, Iloilo City, Dumangas (Panay); Bacolod/adjacent Negros cities; additional Cebu–Bohol–Siquijor towns.

**Interpretation**
At IV, reports are **far from source** and dominated by basin/soil effects (e.g., Iloilo–Panay lowlands, Bacolod plain). Annuli become inconsistent; the inversion follows the **centroid of far-field reports** rather than the fault.

**After filtering / result**

- **Recalculated Epicenter: 10.9995° N, 122.7799° E**
- **Distance from PHIVOLCS epicenter: ≈ 121.061 km**

**Assessment**
IV is **not** a reliable constraint on epicenter here; it mainly delineates the **outer strong-felt envelope**.

**INTENSITY III (Weak)**

**Provinces covered:** Visayas–Mindanao wide (e.g., Leyte, Samar, Bicol entries also appear in some lists)
**Towns included (sample):** Butuan, Tacloban, Catbalogan, Legazpi/Masbate corridor, plus mid-distance Visayas points.

**Interpretation**
III maps **far-field perceptibility** across hundreds of kilometers. Long-period content, building response, and soil resonance dominate. After removing the worst scatterers, a mid-range arc provides a limited directional hint.

**After filtering / result**

- **Recalculated Epicenter: 10.3701° N, 122.9827° E**
- **Distance from PHIVOLCS epicenter: ≈ 47.795 km**

**Assessment**
Surprisingly **better** than IV in this dataset because resampling left a mid-distance cluster in the Visayas core. Still, III is **not** recommended for precise location—more a regional check.

**INTENSITY II (Slightly Felt)**

**Provinces covered:** Bicol, Eastern Samar, Aklan, Camiguin, Bukidnon, broader Visayas–Mindanao fringe
**Towns included (sample):** Sorsogon City, Borongan, Lezo/Aklan; Camiguin/Bukidnon entries; scattered far-field municipalities.

**Interpretation**
II is **perception-dominated** (few felt reports indoors at rest). Radii have the largest spread and mainly trace the maximum reach of long-range waves through island/sea paths.

**After filtering / result**

- **Recalculated Epicenter: 11.0314° N, 123.4396° E**
- **Distance from PHIVOLCS epicenter: ≈ 122.486 km**

**Assessment**
Unsuitable for epicenter location; useful for **felt-area extent** and for confirming attenuation trends only.

## CONSOLIDATED PHYSICAL INTERPRETATION

**Macroseismic Field Behavior and Epicentral Characteristics**
The Negros 2012 event is shaped by a **blind reverse fault** beneath northeastern Negros and a **multi-island geometry** (Negros–Cebu–Bohol–Siquijor):

- **Highest intensity (VII):** Severe shaking along a **single coastline strip** gives **poor azimuthal coverage**, producing very large epicentral error despite near-source damage.
- **Intermediate bands (VI–V):** Add **multi-azimuth** information from Negros Occidental, Cebu, Bohol, and Siquijor. Accuracy improves to **≈ 53 km (VI)** and **≈ 75 km (V)**—reasonable given cross-island path and site variability.

- **Outer bands (IV–II):** Governed by **basin amplification** (Iloilo–Panay, Bacolod plain), long-period propagation, and perception. They map the **felt envelope** but do not localize the source (errors ≈ 48–122 km; IV worst at ≈ 121 km).

**Trends and Transitions**

- **Spatial contraction with intensity:** II–III form a wide Visayas–Mindanao halo → IV/V constrict toward Central Visayas → VI/VII collapse onto the NE Negros coast.
- **Geology controls at mid/low intensities:** Basins and coastal plains inflate IV–V; reclaimed/soft soils around Cebu City elevate V.
- **Geometry dominates high-intensity performance:** Without inland or opposite-coast VII points, triangulation fails—even though damage is clearly near source.

**Overall conclusion**

For the **Negros Oriental 2012** earthquake, intensity-based epicenter estimation behaves exactly as seismology predicts in an archipelagic, basin-rich setting: **excellent physical signal but geometry-limited at VII**, **useful and moderate accuracy at VI–V**, and **perceptibility-only** information at IV–II.